CHANGE NOTICES ARE NOT CUMULATIVE AND SHALL BE RETAINED UNTIL SUCH TIME AS THE ENTIRE STANDARD IS REVISED. RETAIN CHANGE NOTICE 1.

FED. TEST METHOD STD. NO. 228
April 14, 1967
CHANGE NOTICE 2
17 October 1985

FEDERAL TEST METHOD STANDARD

CABLE AND WIRE, INSULATED; METHODS OF TESTING

The following changes which form a part of FED. TEST METHOD STD. NO. 228, dated April 14, 1967, are approved by the Commissioner, Federal Supply Services, General Services Administration, for use by all Federal agencies.

REMOVE: Sections 2, 3, and 4, of January 4, 1984.

ADD: REVISED sections 2, 3, and 4.

ADD: NEW Standard Test Method 2231.

REMOVE: Standard Test Methods 6021 and 6211 of April 14, 1967.

ADD: REVISED Standard Test Methods 6021.1 and 6211.1.

MILITARY INTERESTS:

Custodians

Army - CR
Navy - EC
Air Force - 85

Review Activities

Army - AR, AT
Navy - EC
Air Force - 11, 14, 15, 80, 85, 99
DLA - IS

User Activities

Navy - YD, MC, SH

Agent

DLA - ES

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - CR

DOD project 6145-0888

RETAIN THIS CHANGE NOTICE AND INSERT BEFORE THE TABLE OF CONTENTS OF THIS STANDARD

FSC 6145

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INSULATION STRIPPABILITY

1. Scope.

1.1 This method is intended for use in determining the strippability (adhesion between insulation and conductor) of insulated conductors.

2. Specimen.

2.1 The specimen shall be free of splices, surface cracks, or other visible defects and shall be prepared as shown on figure 1. Preparation shall be done carefully.

3. Apparatus.

3.1 The apparatus shall consist of a tensile tester and a test fixture such as shown on figure 2.

4. Procedure.

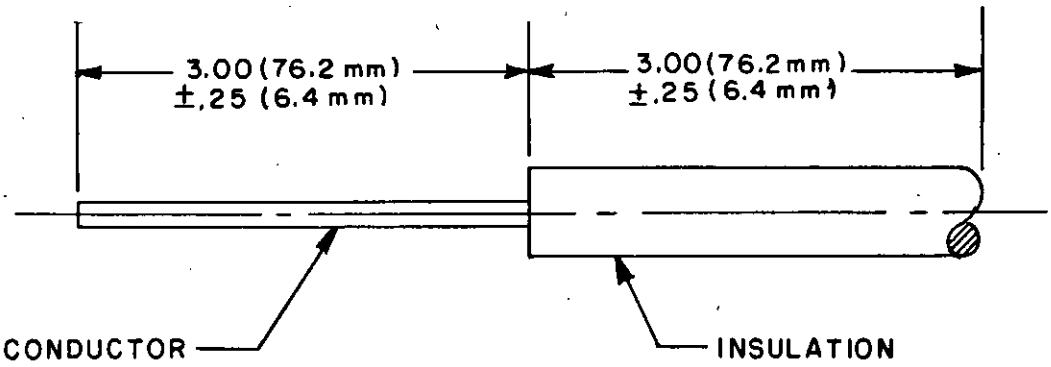
4.1 The adhesion to conductors test shall be performed with a tensile tester and a test fixture such as shown on figure 2. The diameter of the hole in the test plate shall be .004 ±.001 inch larger than the diameter of the applicable conductor. The conductor extending through the test plate hole shall be pulled with a constant rate of 0.2 - 0.5 inch per minute. Avoid sudden pulls and jerking. Conductor adhesion shall be defined as the highest tensile tester reading obtained when the conductor-to-insulation bond is broken. In performing this test, physical handling of the specimen shall be kept to a minimum to avoid specimen degradation.

5. Results.

5.1 The adhesion to conductor (strippability) requirement, as noted by the reading on the tensile tester, shall meet the specified value.

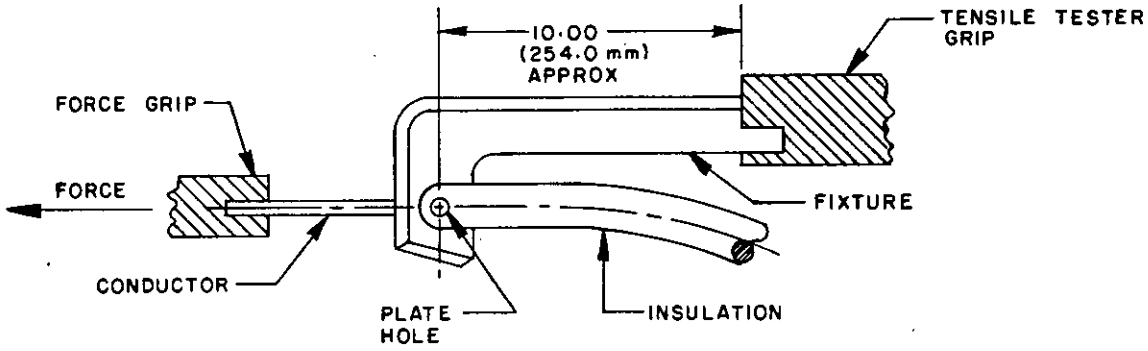
5.2 Where the insulation is stripped, there shall be no evidence of conductor damage and only a trace of insulation or adhesive remaining which can be easily removed by peeling.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Stripping dimensions.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Typical test fixture.

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RESISTANCE, ELECTRICAL CONDUCTOR

1. Scope.

1.1 This method is intended for use in determining the electrical resistance of solid or stranded conductors.

2. Specimen.

2.1 The specimen shall be free of splices, surface cracks, or other visible defects and shall be at least 36 inches in length.

3. Apparatus.

3.1 Resistance shall be measured with a Kelvin bridge, or a suitable four-terminal test method alternate if the resistance is below one ohm. Measurement accuracy shall be within 0.1 percent of reading.

3.2 Resistance may be measured with a Wheatstone bridge, or a suitable alternate if the resistance is above one ohm. Measurement accuracy shall be within one percent of reading.

3.3 Temperature measuring equipment shall be used that will determine the temperature of the conductor to within ±0.5°C (0.9°F).

3.4 A rule shall be used to measure the length of the specimen to an accuracy of ±0.2 percent.

4.0 Procedure.

4.1 If a Wheatstone bridge is used, the resistance of the leads connecting the bridge to the specimen shall be obtained with the leads short-circuited on themselves and the result subtracted from the measured resistance of the specimen.

4.2 If a Kelvin bridge is used, separate current and potential leads shall be used. The current leads shall be attached in such a manner as to give assured contact with all the strands of the conductor. The potential lead clamps shall be such as to encircle the conductor and of small enough width so that the tested length can be assured within the ±0.2 percent accuracy. The distance between the current and potential lead contact shall be greater than three times the diameter of the specimen.

4.3 The test specimen should be allowed to come to the same temperature as the surrounding medium to ensure a correct reading.

4.4 In all resistance measurements, the measuring current raises the temperature of the conductor. Therefore, care should be taken to keep the magnitude of current low, and the time of its use short enough so that the change in resistance cannot be detected.

4.5 To eliminate errors due to contact potential, two readings, one direct and one with current reversed, may be taken in direct succession and the results averaged.

4.6 The resistance shall be measured and recorded, then converted to ohms per unit length. The temperature, conductor length, and conductor size shall be measured and recorded.

5.0 Temperature correction.

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5.1 When the measurement is made at any other than a reference temperature, the resistance may be corrected for moderate temperature differences to what it would be at the reference temperature as follows:

$$R_T = \frac{R_t}{1 + \alpha_T (t-T)}$$

Where:

R_T = Resistance at reference temperature T.
R_t = Resistance as measured at temperature t.
α_T = Known or given temperature coeffficient of resistance of the specimen being measured at reference temperature T.
T = Reference temperature.
t = Temperature at which measurement is made.

NOTE: The parameter α_T, in the above equation, varies with conductivity and temperature. For copper of 100 percent conductivity and a reference temperature of 20°C, its value is 0.00393. Values at other conductivities and temperatures will be found in NBS Handbook 100 of the National Bureau of Standards. Table 2 lists temperature coefficients for the common electrical conductor materials.

6.0 Report.

6.1 The report shall include:

6.2 Specimen length.

6.3 Conductor size.

6.4 Resistance measured.

6.5 Test temperature.

6.6 Correction calculation (if applicable).

6.7 Method of measurement.

6.8 Calculation to appropriate unit length resistance.

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INSULATION DEFECTS, SPARK TEST

1. Scope.

1.1 This method is intended for use in detecting defects in the insulation of insulated wires and cables. Presence of a weak spot in the insulation results in a breakdown at that spot. When breakdown occurs, the spark test equipment is arranged to either automatically stop the coiling equipment or produce a visible or an audible signal. From the manufacturer's viewpoint, this method is preferred to the immersion test method as it constitutes a continuous test that can be used at any time after application of the insulation. It is, in effect, instantaneous, whereas the standard voltage withstand test and insulation resistance test require the immersion of the wire in water for a minimum period of 12 hours. It is not intended that the spark test completely replace the voltage withstand test and insulation resistance test.

2. Specimen.

2.1 Unless otherwise specified, the specimen shall be the entire length of wire or cable offered for inspection.

3. Apparatus.

3.1 The apparatus shall be as follows:

3.1.1 Spark tester providing an essentially sinusoidal voltage.

3.1.1.1 A transformer of sufficient capacity to maintain the test voltage specified in the detail specification under all normal conditions of leakage current.

3.1.1.1.1 The core of the transformer and one end of the secondary winding shall be connected to ground.

3.1.1.2 A voltmeter shall be so located in the circuit that it will indicate at all times the actual test voltage applied.

3.1.1.3 The spark tester shall not be simultaneously connected to more than one electrode described in 3.1.2.

3.1.2 Electrode.

3.1.2.1 An electrode, such as a bead-chain or wire brush, which makes direct mechanical contact with the surface of the insulation undergoing test shall be used. A pipe, coiled spring, or the like shall not be acceptable.

3.1.2.2 If the link or bead-chain type of electrode is used, the bottom of the metal electrode enclosure shall be V-shaped. The chains shall have a length appreciably greater than the depth of the enclosure. The width of the trough shall be approximately 1 1/2 inches greater than the diameter of the largest wire or cable to be tested.

3.1.2.3 If a bead-chain type of electrode is used, the beads shall have a diameter of not greater than 3/16 inch. The longitudinal spacing of the chains shall be not more than 1/2 inch. The transverse spacing of the chains shall be not more than 3/8 inch, except that the spacing may be 1/2 inch if the transverse rows of chain are staggered.

3.1.2.4 The electrode shall be provided with a grounded metallic screen, or the equivalent, as a guard against contact by personnel.

3.1.2.5 The length of electrode shall be sufficient to meet the requirements in 4.3.

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3.1.3 Fault signaling device.

3.1.3.1 A fault signaling device or system shall include a visible signal, a defect recording device, and/or an automatic stop device. The arrangement shall be such that when the fault signal is given, it will be maintained until manually reset.

4. Procedure.

4.1 The spark test shall be conducted as near the end of the manufacturing process as is practicable, preferably as the wire or cable is being cut into shipping lengths.

4.2 The test voltage shall be as specified in the detail specification. The voltmeter shall be connected in the circuit to indicate the actual test potential at all times.

4.3 The conductor or shield of shielded and jacketed cable shall be earth grounded during the spark test. An earth-ground connection shall be made at both the pay-off and take-up reels except that, if the wire is tested for continuity and the conductor is of one integral length, the earth-ground connection need be made at only one point; ie, either the take-up or pay-off reel. In any case, the conductor on a reel at wich an earth-ground connection is made shall be bonded directly to the earth ground on the transformer in the spark tester.

4.4 The length of the electrode is not specified, but the rate of speed at which the wire travels through the electrode shall result in any point on the wire being in contact with the electrode for not less than a total of 18 positive and negative crests of the supply voltage (the equivalent of 9 full cycles of the supply voltage) The maximum acceptable speed of the wire shall be determined by the following formula:

$$\text{Feet per minute} = \frac{5}{9} \times \text{frequency in hertz} \times \text{electrode length in inches}$$

5. Results.

5.1 Unless otherwise specified in the detail specification, the entire delivery of the wire or cable shall be tested.

5.2 Whether the insulation of the entire delivery withstood the specified test voltage without failure shall be recorded.

5.3 The test voltage shall be recorded.

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